

Reverse BIC-cherry

Algorithm 7: Reverse BIC-cherry

Require: properly coloured digraph (G, σ) on L with the sicor-in-hub property, root ρ

Ensure: leaf-coloured network (N, σ) ; $c(a, b)$: node(s) of N that make b a best match of a

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1:  $V(N) \leftarrow L \cup \{\rho\}$ ;  $E(N) \leftarrow \emptyset$ ;  $W \leftarrow \emptyset$ 
2: if  $|L| = 2$  then add edges  $(\rho, x)$  for all  $x \in L$  and return  $(N, \sigma)$ 
3: for all  $\{x, y\} \subseteq L$  with  $(x, y), (y, x) \in E(G)$  do                                ▷ step 1: reciprocal edges
4:   add  $p_{xy}$  with edges  $(\rho, p_{xy}), (p_{xy}, x), (p_{xy}, y)$ ;  $c(x, y), c(y, x) \leftarrow \{p_{xy}\}$ 
5: end for
6: for all  $(x, y) \in E(G)$  with  $(y, x) \notin E(G)$  do                                ▷ step 2: partner of  $x$ 's colour
7:    $Z \leftarrow \{z \in L : \sigma(z) = \sigma(x), (y, z), (z, y) \in E(G)\}$ 
8:   for all  $z \in Z$  do add  $q$  with edges  $(\rho, q), (q, x), (q, p_{yz})$ ;  $c(x, y) \leftarrow c(x, y) \cup \{q\}$ 
9:   if  $Z = \emptyset$  then  $W \leftarrow W \cup \{(x, y)\}$                                 ▷ left for step 3
10: end for
11: while  $W \neq \emptyset$  do                                                        ▷ step 3: search from  $y$ 
12:   for all  $(x, y) \in W$  whose required  $c(y, \cdot)$  exist do
13:     for all  $t \in \text{FINDTARGETS}(G, x, y)$  do add  $r$  with edges  $(\rho, r), (r, x), (r, t)$ ;  $c(x, y) \leftarrow c(x, y) \cup \{r\}$ 
14:      $W \leftarrow W \setminus \{(x, y)\}$ 
15:   end for
16: end while
17: return  $(N, \sigma)$ 
```

Reverse BIC-cherry: step 3

Function FindTargets (step 3 of Algorithm 7)

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1: function FINDTARGETS( $G, x, y$ )
2:   for  $d = 1, 2, \dots$  do                                     ▷ outward from  $y$ , nearest first
3:      $H_d \leftarrow \{w \in L \setminus \{x\} : \sigma(w) = \sigma(x), \text{dist}_G(y, w) = d\}$            ▷ nodes of  $x$ 's colour
4:      $T_d \leftarrow \{c(y, u) : y \rightarrow u \rightarrow \dots \rightarrow w \text{ is a shortest path to some } w \in H_d\}$ 
5:      $T_d^* \leftarrow \{t \in T_d : (x, \ell) \in E(G) \text{ for all } \ell \in L(t) \text{ with } \sigma(\ell) \neq \sigma(x)\}$            ▷ clean cluster
6:     if  $T_d^* \neq \emptyset$  then return  $T_d^*$ 
7:   end for
8:   return  $T_{d_0}$  for the smallest  $d_0$  with  $H_{d_0} \neq \emptyset$            ▷ fallback: nearest
9: end function
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Reverse BIC-cherry: simplified

Algorithm 8: Reverse BIC-cherry, simplified

Require: properly coloured digraph (G, σ) on L with the sicor-in-hub property

Ensure: leaf-coloured network (N, σ)

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1:  $k \leftarrow 0$ ;  $V(N) \leftarrow \{\rho_0\}$ ;  $E(N) \leftarrow \emptyset$ 
2: repeat
3:    $A \leftarrow \{v \in V(G) : N^+(v) = N^-(v) = V(G) \setminus \{v\}\}$ ;  $\text{ATTACH}(A)$ 
4:    $B \leftarrow \{v \in V(G) : N^-(v) = \emptyset, N^+(v) = \{u : \sigma(u) \neq \sigma(v)\}\}$ ;  $\text{ATTACH}(B)$ 
5: until  $A = B = \emptyset$ 
6: if  $E(G) = \emptyset$  then
7:   add edges  $(\rho_k, v)$  for all  $v \in V(G)$ ; remove  $\rho_k$  if it has no children
8: else
9:   add Algorithm 7 on the remaining  $(G, \sigma)$  with root  $\rho_k$  to  $N$ 
10: end if
11: return  $(N, \sigma)$ 

12: procedure  $\text{ATTACH}(S)$ 
13:   if  $S \neq \emptyset$  then
14:     add edges  $(\rho_k, v)$  for all  $v \in S$ ; remove  $S$  from  $G$ 
15:     add  $\rho_{k+1}$  with edge  $(\rho_k, \rho_{k+1})$ ;  $k \leftarrow k + 1$ 
16:   end if
17: end procedure
```

▷ peel off leaves from the top

▷ reciprocal to all

▷ no incoming